Implementing Enterprise Network

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NTI : Network Security

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DEVICE NAME	INTERFACE	IP ADDRESS	SUBNET MASK	DEFAULT GATEWAY	SWITCH PORT
R1	G0/0/0	192.168.1.25	255.255.255.248	N/A	S1 F0/5
	S0/1/0	192.168.1.89	255.255.255.252	N/A	N/A
R2	S0/2/0	192.168.1.90	255.255.255.252	N/A	N/A
	S0/2/1	192.168.1.93	255.255.255.252	N/A	N/A
	S0/1/0	192.168.1.97	255.255.255.252	N/A	N/A
	S0/1/1	192.168.1.101	255.255.255.252	N/A	N/A
R3	G0/0/0	192.168.1.1	255.255.255.240	N/A	S2 F0/7
	S0/1/0	192.168.1.94	255.255.255.252	N/A	N/A
R4	G0/0/0	192.168.1.41	255.255.255.248	N/A	S4 F0/1
	G0/0/1	192.168.1.33	255.255.255.248	N/A	S3 F0/3
	S0/1/0	192.168.1.98	255.255.255.252	N/A	N/A
R5	S0/2/1	192.168.1.102	255.255.255.252	N/A	N/A
	S0/2/0	192.168.1.113	255.255.255.252	N/A	N/A
	S0/1/1	192.168.1.109	255.255.255.252	N/A	N/A
	S0/1/0	192.168.1.105	255.255.255.252	N/A	N/A
R6	G0/0/0.10	192.168.1.57	255.255.255.248	N/A	S5 F0/3
	G0/0/0.99	192.168.1.49	255.255.255.248	N/A	S5 F0/3
	S/1/0	192.168.1.106	255.255.255.252	N/A	N/A
R7	G0/0/0	192.168.1.17	255.255.255.248	N/A	S7 F0/4
	G0/0/1	192.168.1.65	255.255.255.248	N/A	S6 F0/3
	S0/1/0	192.168.1.110	255.255.255.252	N/A	N/A
R8	G0/0/0	192.168.1.81	255.255.255.248	N/A	S9 0/2
	S0/1/0	192.168.1.114	255.255.255.252	N/A	N/A
	S0/1/1	192.168.1.117	255.255.255.252	N/A	N/A
R9	G0/0/0	192.168.1.73	255.255.255.248	N/A	S8 F0/3
	S0/1/0	192.168.1.118	255.255.255.252	N/A	N/A
S1	VLAN 1	192.168.1.26	255.255.255.248	192.168.1.25	N/A
S2	VLAN 1	192.168.1.2	255.255.255.240	192.168.1.1	N/A
S3	VLAN 1	192.168.1.34	255.255.255.248	192.168.1.33	N/A
S4	VLAN 1	192.168.1.42	255.255.255.248	192.168.1.41	N/A
S5	VLAN 99	192.168.1.52	255.255.255.248	192.168.1.49	N/A
S6	VLAN 1	192.168.1.66	255.255.255.248	192.168.1.65	N/A

S7	VLAN 1	192.168.1.18	255.255.255.248	192.168.1.17	N/A
S8	VLAN 1	192.168.1.74	255.255.255.248	192.168.1.73	N/A
S9	VLAN 1	192.168.1.82	255.255.255.252	192.168.1.81	N/A
S10	VLAN 99	192.168.1.51	255.255.255.248	192.168.1.49	N/A
S11	VLAN 99	192.168.1.50	255.255.255.248	192.168.1.49	N/A
PC1	NIC	192.168.1.28	255.255.255.248	192.168.1.25	S1 F0/3
PC2	NIC	192.168.1.27	255.255.255.248	192.168.1.25	S1 F0/2
PC3	NIC	192.168.1.3	255.255.255.240	192.168.1.1	S2 F0/6
PC4	NIC	192.168.1.4	255.255.255.240	192.168.1.1	S2 F0/5
PC5	NIC	192.168.1.7	255.255.255.240	192.168.1.1	S2 F0/2
PC6	NIC	192.168.1.6	255.255.255.240	192.168.1.1	S2 F0/3
PC7	NIC	192.168.1.5	255.255.255.240	192.168.1.1	S2 F0/4
PC8	NIC	192.168.1.8	255.255.255.240	192.168.1.1	S2 F0/1
PC9	NIC	192.168.1.36	255.255.255.248	192.168.1.33	S3 F0/2
PC10	NIC	192.168.1.35	255.255.255.248	192.168.1.33	S3 F0/1
PC11	NIC	192.168.1.44	255.255.255.248	192.168.1.41	S4 F0/2
PC12	NIC	192.168.1.58	255.255.255.248	192.168.1.57	S10 F0/3
PC13	NIC	192.168.1.59	255.255.255.248	192.168.1.57	S11 F0/3
PC14	NIC	192.168.1.19	255.255.255.248	192.168.1.17	S7 F0/3
PC15	NIC	192.168.1.75	255.255.255.248	192.168.1.73	S8 F0/1
PC16	NIC	192.168.1.83	255.255.255.248	192.168.1.81	S9 F0/1
WEB SERVER	NIC	192.168.1.43	255.255.255.248	192.168.1.41	S4 F0/3

NTP SERVER	NIC	192.168.1.68	255.255.255.248	192.168.1.65	S6 F0/2
SYSLOG SERVER	NIC	192.168.1. 67	255.255.255.248	192.168.1.65	S6 F0/1
RADIUS SERVER	NIC	192.168.1.21	255.255.255.248	192.168.1.17	S7 F0/1
TACACS+ SERVER	NIC	192.168.1.20	255.255.255.248	192.168.1.17	S7 F0/2
WEB SERVER 2	NIC	192.168.1.76	255.255.255.248	192.168.1.73	S8 F0/2

I began by determining the required network size and the number of devices. After that, I performed subnetting and assigned IP addresses according to the subnetting plan. I verified LAN connectivity using ping between devices.

Next, I applied the basic configuration on all network devices, including setting hostnames, MOTD banners, usernames, and passwords.

Finally, I enabled OSPF routing, advertised the necessary subnets, and verified that routing was functioning correctly using commands like ping, show ip route, and show ip ospf neighbor.

After completing the basic setup and verifying connectivity, I proceeded to implement some security concepts :

1-Enable SSH on all network devices

```
R1#sho ip ssh
SSH Enabled - version 1.99
Authentication timeout: 120 secs; Authentication retries: 3
R1#
R1#show running-config | section line vty
line vty 0 4
 login authentication default
 transport input ssh
line vty 5 15
 login authentication default
 transport input ssh
R1#
```

2-Enable OSPF authentication on all routers

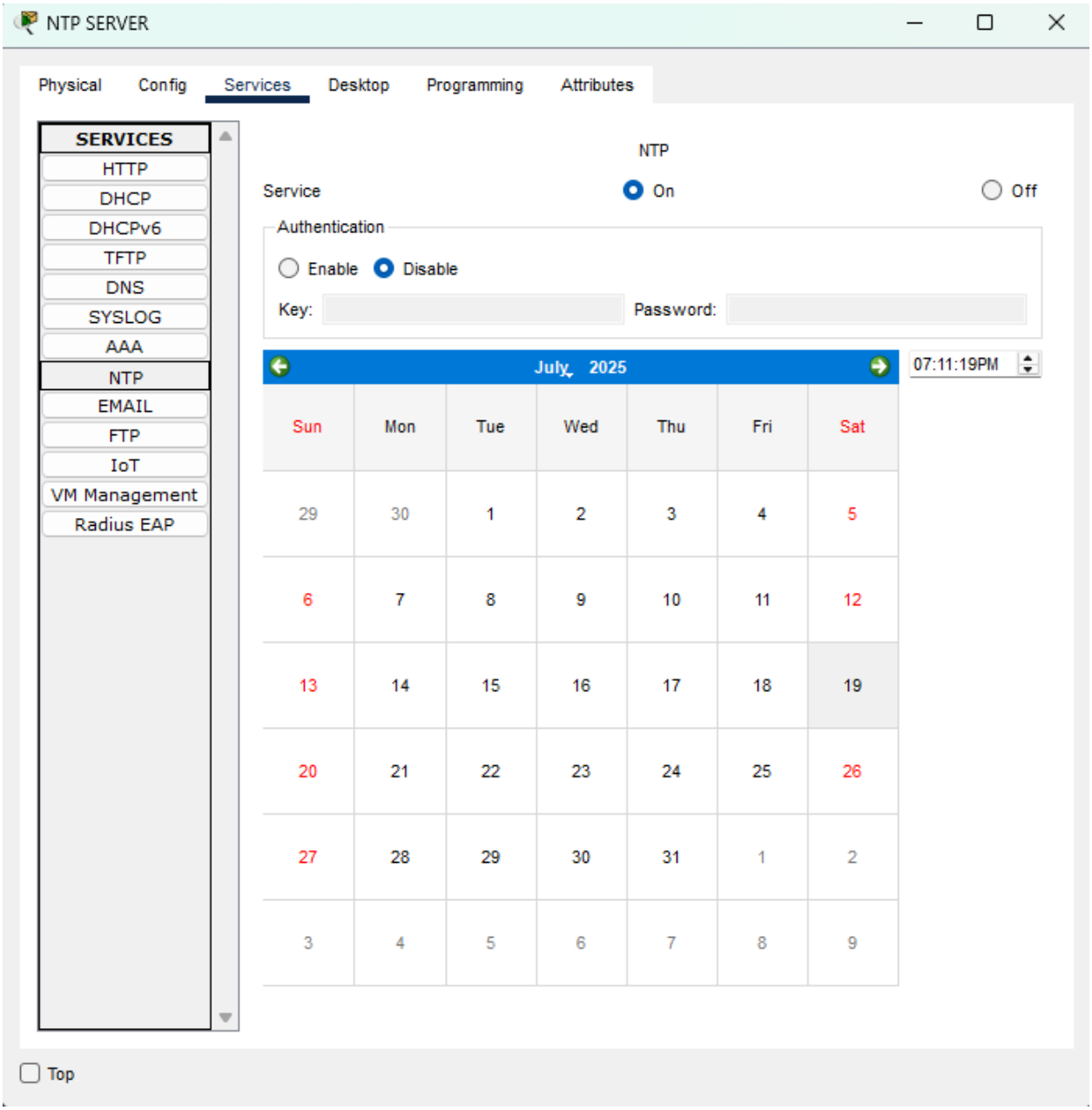
```
interface Serial0/1/0
 ip address 192.168.1.105 255.255.255.252
 ip ospf message-digest-key 1 md5 cisco
!
interface Serial0/1/1
 ip address 192.168.1.109 255.255.255.252
 ip ospf message-digest-key 1 md5 cisco
!
interface Serial0/2/0
 ip address 192.168.1.113 255.255.255.252
 ip ospf message-digest-key 1 md5 cisco
!
interface Serial0/2/1
 ip address 192.168.1.102 255.255.255.252
 ip ospf message-digest-key 1 md5 cisco
 clock rate 2000000
!
```

3-Create views on router 7

```
parser view cisco
 secret 5 $1$mERr$hx5rVt7rPNoS4wqbXKX7m0
 commands exec include configure
 commands exec include all configure terminal
 commands exec include all show
!
parser view cisco1
 secret 5 $1$mERr$hx5rVt7rPNoS4wqbXKX7m0
 commands exec include all show
!
```

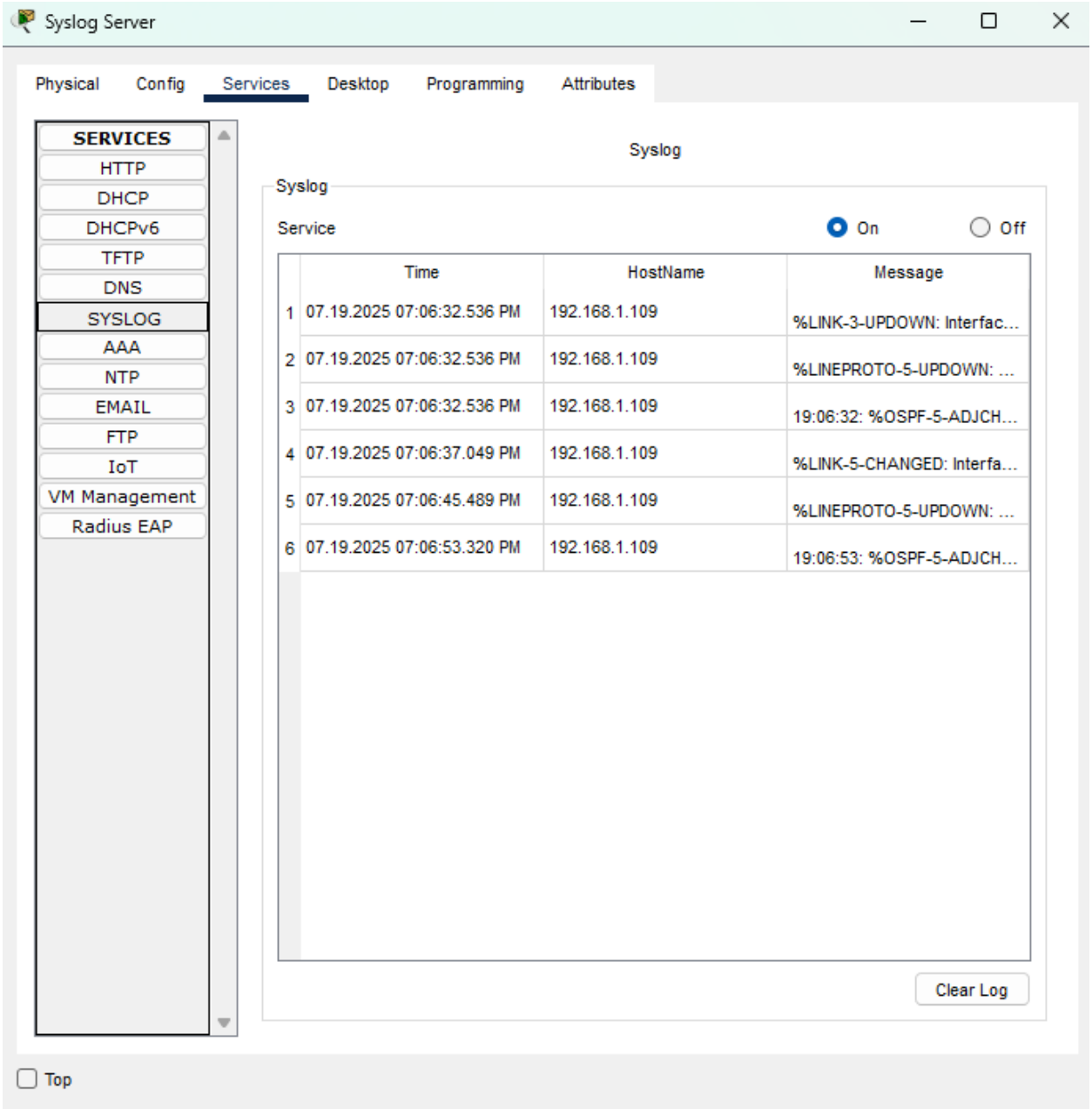
4-Enabling NTP server and all network devices connected to it

```
.
ntp server 192.168.1.68
!
end
```



5-Enabling SYS-LOG server and all network devices connected to it

```
logging 192.168.1.67
```



6-Enabling RADIUS and TACACS+ servers and make all network devices connected to it as the RADIUS is the main server and the TACACS+ server is the back up then the local authentication

```
aaa authentication login default group radius group tacacs+ local
```

RADUIS Server

Physical | Config | Services | Desktop | Programming | Attributes

SERVICES

- HTTP
- DHCP
- DHCPv6
- TFTP
- DNS
- SYSLOG
- AAA
- NTP
- EMAIL
- FTP
- IoT
- VM Management
- Radius EAP

AAA

Service: On / Off — Radius Port: 1645

Network Configuration

Client Name: Client IP:

Secret: ServerType: Radius

	Client Name	Client IP	Server Type	Key
1	R6	192.168.1.106	Radius	cisco
2	R8	192.168.1.114	Radius	cisco
3	R9	192.168.1.118	Radius	cisco
4	R7	192.168.1.17	Radius	cisco
5	S7	192.168.1.18	Radius	cisco
6	S1	192.168.1.26	Radius	cisco

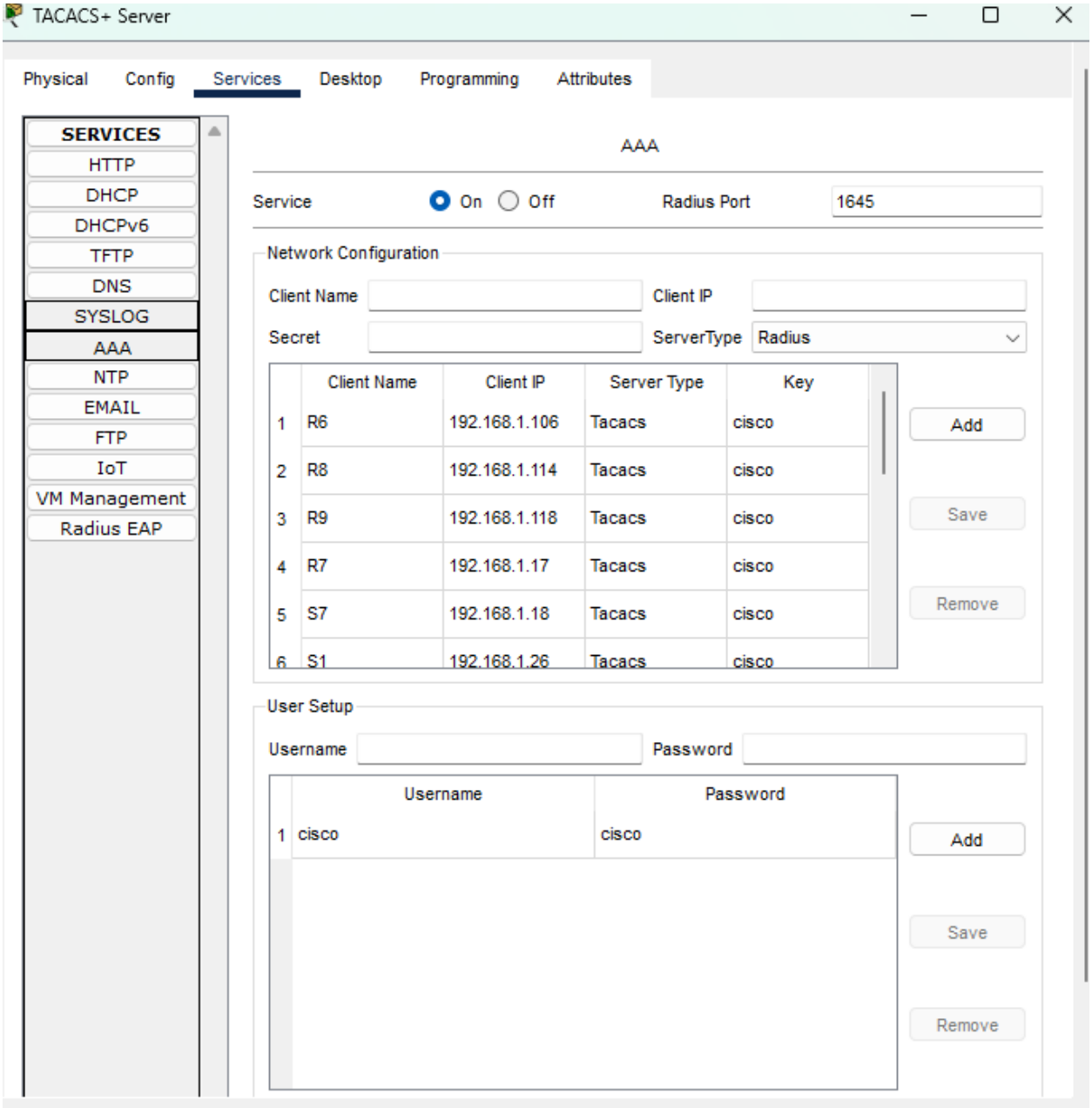
Add | Save | Remove

User Setup

Username: Password:

	Username	Password
1	cisco	cisco

Add | Save | Remove



Except router 2 and router 5 they have their local AAA authentication

```
!
aaa authentication login default local
!
```

7-ACL on the blue network (Router 3)

a-Prevent private (except 192.168.1.0/24) ,multicast , loop back and DHCP connection from outside the network

```
Extended IP access list 100
    10 deny ip host 0.0.0.0 any
    20 deny ip 10.0.0.0 0.255.255.255 any
    30 deny ip 127.0.0.0 0.255.255.255 any
    40 deny ip 172.16.0.0 0.15.255.255 any
    50 deny ip 224.0.0.0 15.255.255.255 any
    60 deny ip host 255.255.255.255 any
    70 permit ip any any
```

b-Permit the http requests from the inside network to the outside network and permit replies for it ; prevent outside requests to http

```
Extended IP access list 120
    10 permit tcp 192.168.1.0 0.0.0.15 any eq www
    20 permit tcp any 192.168.1.0 0.0.0.15 established
    30 deny tcp any 192.168.1.0 0.0.0.15 eq www
    40 permit ip any any
```

c-Prevent any ICMP messages from outside network (echo or echo reply)

```
Extended IP access list 140
    10 deny icmp any 192.168.1.0 0.0.0.15 echo
    20 deny icmp any 192.168.1.0 0.0.0.15 echo-reply
    30 permit ip any any (434 match(es))
```

8-ZPF (Router 4)

8.1 Create the security zones

```
zone security INSIDE
zone security OUTSIDE
zone security DMZ
```

8.2 Assign interfaces to zones

```
interface GigabitEthernet0/0/0
 ip address 192.168.1.41 255.255.255.248
 zone-member security DMZ
 ip ospf message-digest-key 1 md5 cisco
 duplex auto
 speed auto
!
interface GigabitEthernet0/0/1
 ip address 192.168.1.33 255.255.255.248
 zone-member security INSIDE
 ip ospf message-digest-key 1 md5 cisco
 duplex auto
 speed auto
!
interface GigabitEthernet0/0/2
 no ip address
 duplex auto
 speed auto
 shutdown
!
interface Serial0/1/0
 ip address 192.168.1.98 255.255.255.252
 zone-member security OUTSIDE
 ip ospf message-digest-key 1 md5 cisco
!
```

8.3 Create the class map (Allowed traffic)

```
class-map type inspect match-any czpf
 match access-group 101
 match protocol http
 match protocol https
 match protocol ssh
 match protocol icmp
 match protocol ntp
 match protocol syslog
!
```

8.4 Create two policy maps

```
policy-map type inspect iozpf
 class type inspect czpf
  inspect
!
policy-map type inspect odzpf
 class type inspect czpf
  inspect
!
```

8.5 Apply zone pairs

```
zone-pair security io source INSIDE destination OUTSIDE
 service-policy type inspect iozpf
zone-pair security od source OUTSIDE destination DMZ
 service-policy type inspect odzpf
!
```

9-Layer 2 Secruity (Yellow netwrok)

a-Applying STP port fast on interface fa0/3 on both switches 10 and 11

```
 spanning-tree portfast
!
```

b-Applying port security on interface fa0/3 on both switches 10 and 11

```
switchport mode access
switchport port-security
switchport port-security mac-address sticky
switchport port-security mac-address sticky 00E0.8FA0.4339
```

```
S11#sho port-security interface fa0/3
Port Security              : Enabled
Port Status                : Secure-up
Violation Mode             : Shutdown
Aging Time                 : 0 mins
Aging Type                 : Absolute
SecureStatic Address Aging : Disabled
Maximum MAC Addresses      : 1
Total MAC Addresses        : 1
Configured MAC Addresses   : 0
Sticky MAC Addresses       : 1
Last Source Address:Vlan   : 00E0.8FA0.4339:10
Security Violation Count   : 0
```

10-VPN (Router 1 and Router 9)

10.1 Create ACL on both sides to permit traffic between them

```
access-list 100 permit ip 192.168.1.24 0.0.0.7 192.168.1.72 0.0.0.7
```

```
access-list 100 permit ip 192.168.1.72 0.0.0.7 192.168.1.24 0.0.0.7
!
```

10.2 Create ISAKMP policy (phase 1 tunnel) both sides

```
crypto isakmp policy 10
 encr aes 256
 authentication pre-share
 group 5
 lifetime 10000
!
crypto isakmp key cisco address 192.168.1.118
!
!
```

```
crypto isakmp policy 10
 encr aes 256
 authentication pre-share
 group 5
 lifetime 10000
!
crypto isakmp key cisco address 192.168.1.89
!
!
```

10.3 Create IPSec policy (phase 2 tunnel) both sides

```
crypto ipsec transform-set R1-R9 esp-aes esp-sha-hmac
```

10.4 Create the crypto map on both sides

```
crypto map R1-R9-Map 10 ipsec-isakmp
 set peer 192.168.1.118
 set pfs group5
 set security-association lifetime seconds 5000
 set transform-set R1-R9
 match address 100
!
!

crypto map R1-R9-Map 10 ipsec-isakmp
 set peer 192.168.1.89
 set pfs group5
 set security-association lifetime seconds 5000
 set transform-set R1-R9
 match address 100
!
!
```

10.5 Apply on interfaces

```
interface Serial0/1/0
 ip address 192.168.1.118 255.255.255.252
 ip ospf message-digest-key 1 md5 cisco
 clock rate 2000000
 crypto map R1-R9-Map
!

interface Serial0/1/0
 ip address 192.168.1.89 255.255.255.252
 ip ospf message-digest-key 1 md5 cisco
 crypto map R1-R9-Map
!
```

10.6 Verify (trace route)

```
C:\>tracert 192.168.1.27

Tracing route to 192.168.1.27 over a maximum of 30 hops:

  1   0 ms      6 ms      0 ms      192.168.1.73
  2   34 ms     9 ms      20 ms     192.168.1.89
  3   54 ms     42 ms     3 ms      192.168.1.27

Trace complete.

C:\>
```